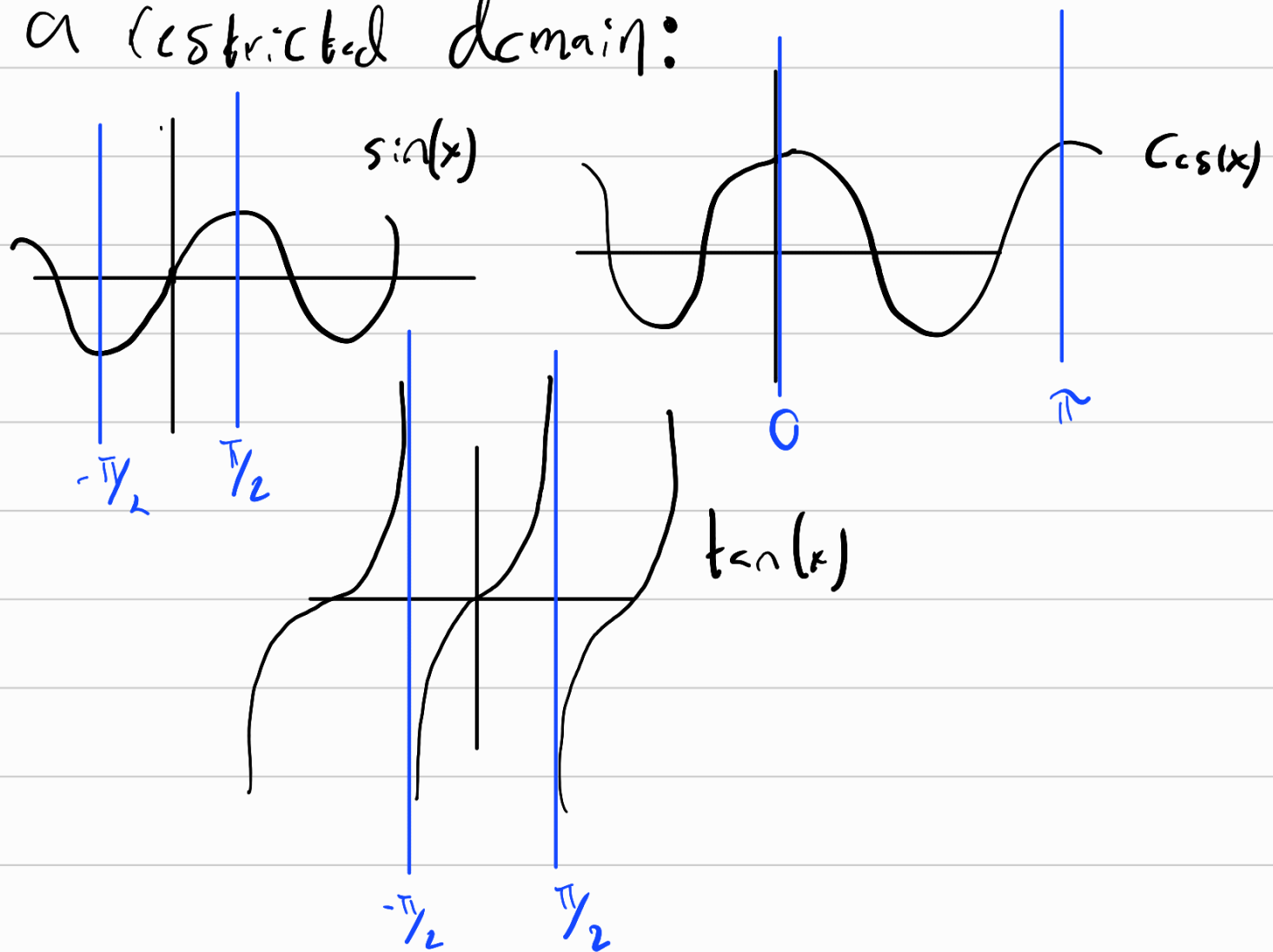


functions like $\sin(x)$, $\cos(x)$, $\tan(x)$ are not one-to-one on their entire domain, So we define their inverse functions on a restricted domain:



This gives the inverse functions

	Domain	Range
$\arcsin(x)$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos(x)$	$[-1, 1]$	$[0, \pi]$
$\arctan(x)$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

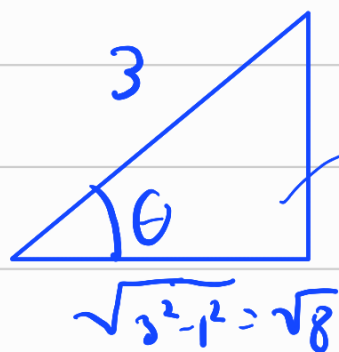
Because of the domain restriction, we have issues like

$$\sin(\arcsin x) = x \quad \text{for } x \in [-1, 1]$$

$$\text{but } \arcsin(\sin x) = x \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

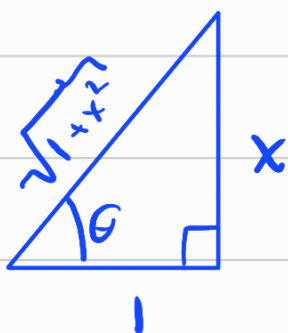
even though domain of $\sin(x)$ is $\mathbb{R}$, we have to restrict it.

Ex: $\tan(\arcsin(\frac{1}{3})) = \frac{1}{2\sqrt{2}}$



$$\tan \theta = \frac{1}{\sqrt{8}} = \boxed{\frac{1}{2\sqrt{2}}}$$

$$\cos(\arctan(x)) = ?$$



$$\cos(\theta) = \boxed{\frac{1}{\sqrt{1+x^2}}}$$

we get the following derivatives

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\operatorname{arccsc} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}, \quad \frac{d}{dx}(\operatorname{arccot} x) = \frac{-1}{1+x^2}$$

Pf/ let $y = \arcsin(x)$, then $\sin(y) = x$

from implicit derivatives

$$\sin(y) = x \rightarrow y' \cos(y) = 1$$

$$y' = \frac{1}{\cos(y)}$$

Recall, $\cos(y) = \sqrt{1-\sin^2(y)}$, so

$$y' = \frac{1}{\sqrt{1-\sin^2(\arcsin x)}} = \boxed{\frac{1}{\sqrt{1-x^2}}}$$

Integration:

Working backwards from above we acquire more general integral formulas. The two most useful being

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$$

$$\int \frac{dx}{1+x^2} = \arctan(x) + C$$

Ex: $\int_0^{1/4} \frac{dx}{\sqrt{1-4x^2}} = \int_0^{1/4} \frac{dx}{\sqrt{1-(2x)^2}}$

$u = 2x, du = 2dx$ then the integral is

$$\frac{1}{2} \int_0^{1/2} \frac{du}{\sqrt{1-u^2}} = \frac{1}{2} \arcsin(u) \Big|_0^{1/2}$$

$$= \frac{1}{2} \underbrace{\arcsin\left(\frac{1}{2}\right)}_{\pi/6} - \frac{1}{2} \underbrace{\arcsin(0)}_0$$

$$= \frac{\pi}{12}$$

Ex:

$$\int \frac{dx}{x^2 + a^2} = \int \frac{dx}{a^2 \left(\left(\frac{x}{a} \right)^2 + 1 \right)}$$

$$= \frac{1}{a^2} \int \frac{dx}{\left(\frac{x}{a} \right)^2 + 1} \quad \begin{array}{l} u = \frac{x}{a} \\ du = \frac{1}{a} dx \end{array}$$

$$= \frac{1}{a} \int \frac{du}{u^2 + 1}$$

$$= \frac{1}{a} \arctan u + C$$

$$= \frac{1}{a} \arctan \left(\frac{x}{a} \right) + C$$